Arduino IDE Setup Guide

FastLineFollower — ESP32 NodeMCU Line Follower Robot

Step 1 — Install Arduino IDE

Download and install **Arduino IDE 2.3.10 or later** from the official site:

```
https://www.arduino.cc/en/software
```

Step 2 — Install the CP210x USB Driver

The ESP32 NodeMCU uses a **CP2102 USB-to-UART chip**. Without the driver, no COM port will appear in the IDE.

How to install:

1. Go to the Silicon Labs driver page:

```
https://www.silabs.com/software-and-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads
```

- 2. Download **CP210x Universal Windows Driver** (v11.6.0 or later).
- 3. Extract the zip file.
- 4. Right-click **CP210xVCPInstaller_x64.exe** → **Run as administrator**.
- 5. Plug in the ESP32 via USB.
- 6. Open **Device Manager** → **Ports (COM & LPT)**. You should see **"Silicon Labs CP210x USB to UART Bridge (COMx)"**. **Write down the COM number** (e.g. COM3, COM8).

Step 3 — Install the ESP32 Board Package

■■ There are two different "esp32" packages. You need the correct one, or the code will not compile.

3a. Add the board URL

Open **File** → **Preferences** (Mac: Arduino IDE → Settings).

Paste this into the "**Additional boards manager URLs**" field:

```
https://espressif.github.io/arduino-esp32/package_esp32_index.json
```

Click **OK**.

3b. Install the package

- 1. Open **Boards Manager** (left sidebar icon that looks like a board).
- 2. Search for **"esp32"**.
- 3. Install **"esp32 by Espressif Systems"** — the latest version.

■ Correct: "esp32 by Espressif Systems"

■ Wrong: "Arduino ESP32 Boards" (that is the Nano ESP32 package — do NOT use it)

Step 4 — Select the Board and Port

1. **Tools** → **Board** → **esp32** → **ESP32 Dev Module**

Make sure the board is under the section named **esp32** (by Espressif), not “Arduino ESP32 Boards”.

2. **Tools** → **Port** → **COMx** (the Silicon Labs CP210x port from Step 2).

Step 5 — Upload the Sketch

- 1. Open **FastLineFollower.ino**.
- 2. Click **Verify** (✓) — should compile cleanly, about 21% flash usage.
- 3. Click **Upload** (→) — you should see **“Hash of data verified”** in the output.

Step 6 — Verify It Works

1. Open **Tools** → **Serial Monitor**, set baud rate to **115200**.

2. Press the **RST** button on the ESP32.

3. You should see this output:

```
MECC 2026 FAST LINE FOLLOWER
Ready.
Press BOOT to start/stop.
Kp=40.0 Kd=60.0 base=240 slowK=20.0 min=120 pivot=180
boxKick=80.0 turnSpeed=170 turnMs=450
Sensors Lbox:0 line:010 Rbox:0
```

Sensor Layout

Three sensors in the middle follow the line; the two on the outside watch for the black boxes on the track.

Sensor	GPIO	Job
IR1	23	Left box sensor — sees a box, robot steers left
IR2	22	Line following (middle left)
IR3	21	Line following (centre, the main sensor)
IR4	19	Line following (middle right)
IR5	18	Right box sensor — sees a box, robot steers right

• All three middle sensors dark (thick box) → the robot turns.
• Otherwise → normal PD line following.

Troubleshooting

Symptom	Fix
No COM port in IDE	Driver not installed, wrong USB cable (must be data cable, not charge-only), or try a different USB port.
'ledcAttach' was not declared	Wrong board package installed. Uninstall “Arduino ESP32 Boards” in Boards Manager. Install “esp32 by Espressif Systems” instead.
dfu-util: No DFU capable USB device	Wrong board selected in Tools → Board. Change to **ESP32 Dev Module** under the “esp32” section.

Upload fails intermittently	Press and hold the **BOOT** button on the ESP32 while clicking Upload. Release when “Connecting…” appears.